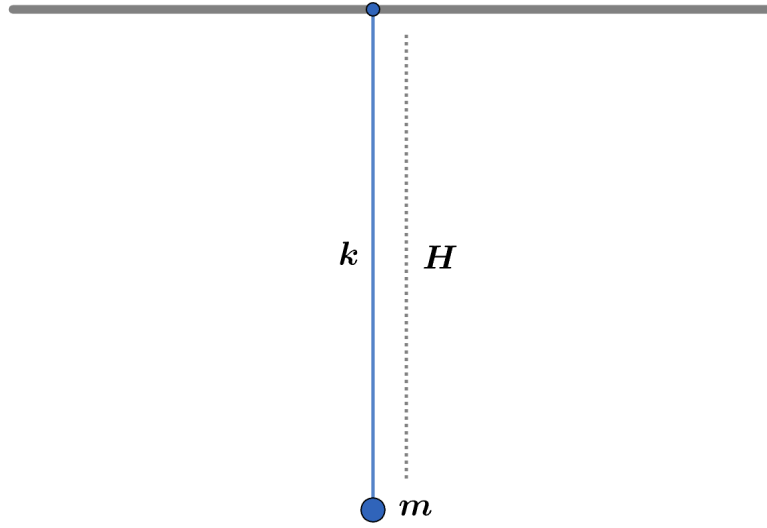


2013 F=ma Exam: Problem 23

Kevin S. Huang



By conservation of energy, the gravitational potential energy of the man is converted into spring potential energy at the bottom of his fall:

$$mgH = \frac{1}{2}kh^2$$

This is also when the tension in the spring is maximized so

$$T = kh = 4mg$$

$$h = \frac{4mg}{k}$$

Substituting this into the first equation,

$$mgH = \frac{1}{2}k \left(\frac{4mg}{k} \right)^2 = \frac{8m^2g^2}{k}$$

Thus,

$$k = \frac{8mg}{H}$$

so the answer is E.